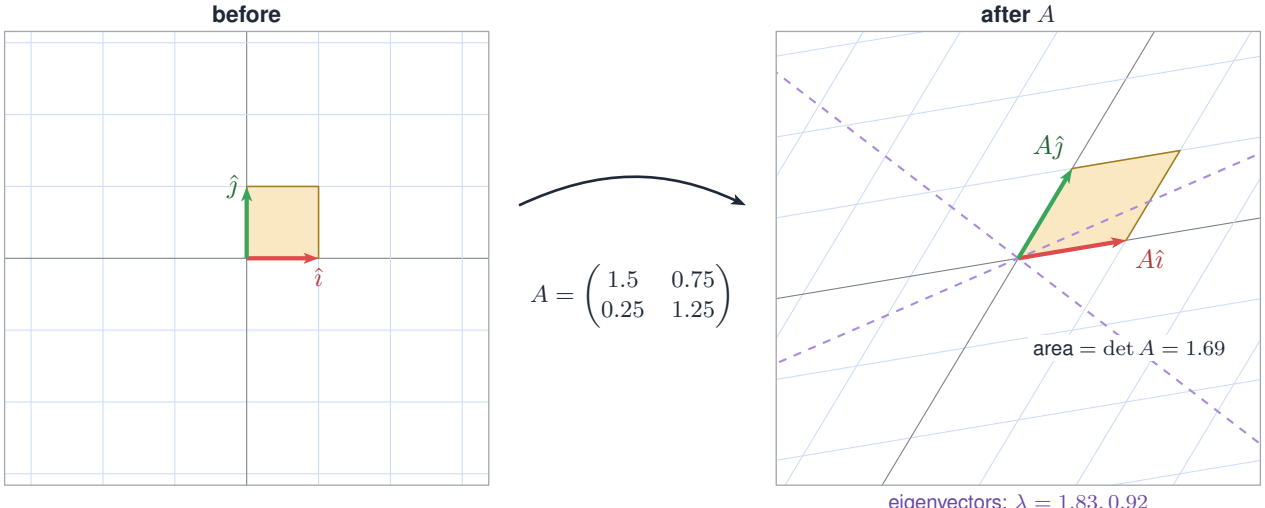

before
$\hat{\jmath}$
$\hat{\imath}$
$A = \begin{pmatrix} 1.5 & 0.75 \\ 0.25 & 1.25 \end{pmatrix}$
after A
$A\hat{\jmath}$
$A\hat{\imath}$
area $= \det A = 1.69$
eigenvectors: $\lambda = 1.83, 0.92$